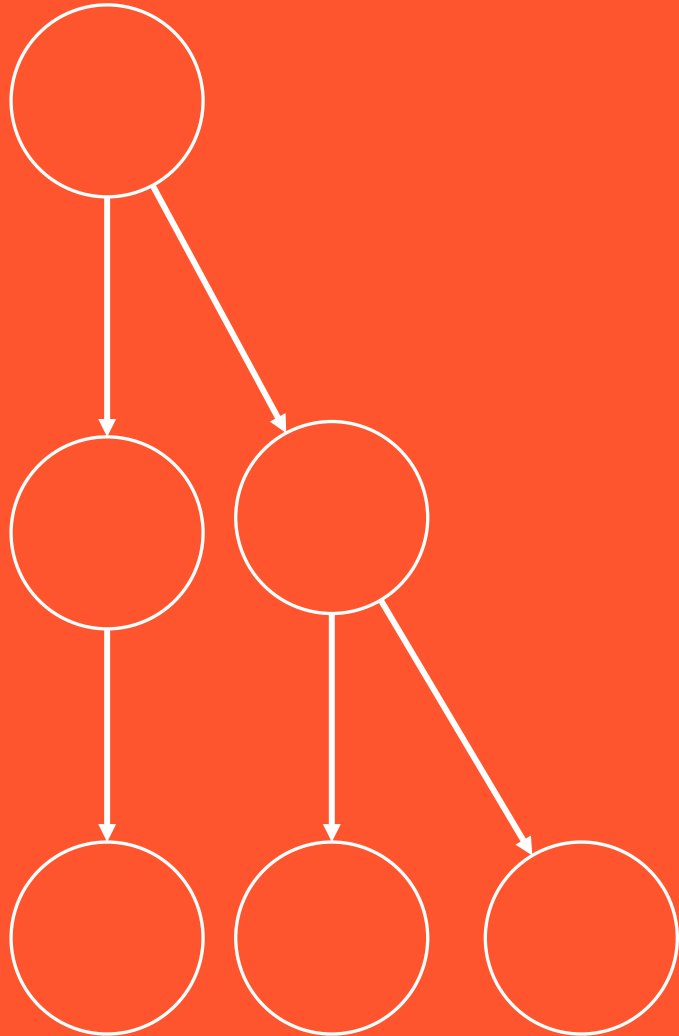


# BCOG 100

Vertebrate Neural Architecture  
Lecture 6.3 — The Vertebrate Brain, and How It Is Built



UNIVERSITY OF  
**ILLINOIS**  
URBANA - CHAMPAIGN

# To Start: Why Release Rather Than Switch On?

In your group, three minutes.

In the last lecture, choosing an action turned out to be *releasing* it, not switching it on.

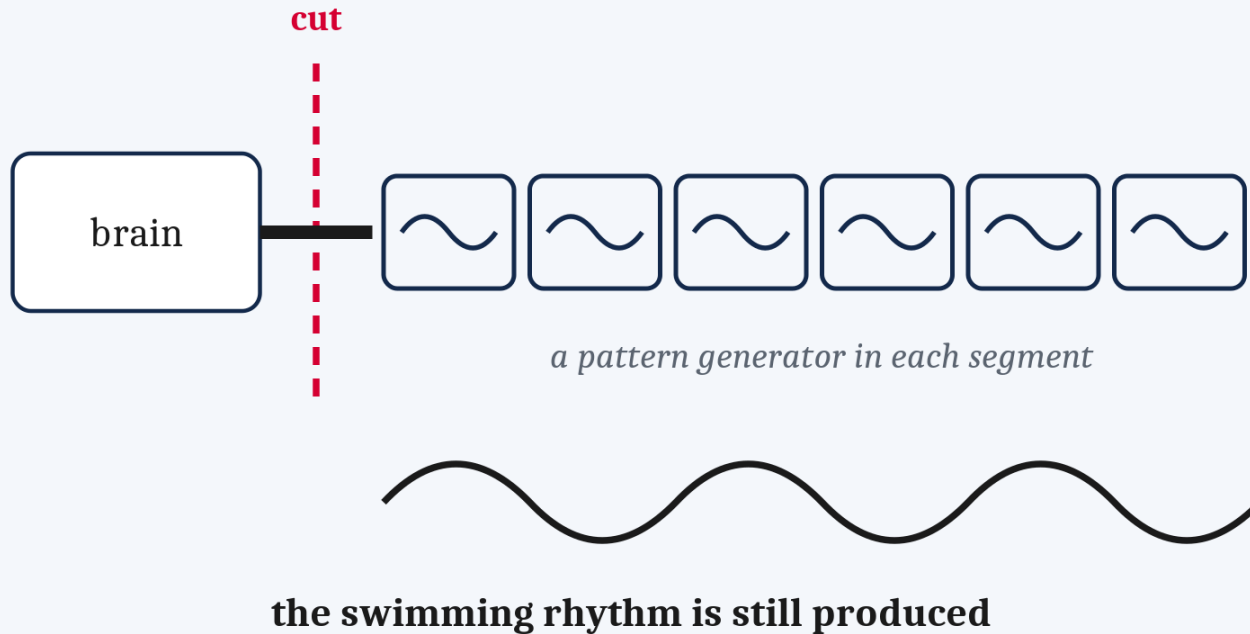
What does that mean, in a sentence?

And why is there no walking organ, no eating organ, no talking organ?

# Hindbrain and Spinal Cord: Running the Body

The oldest-looking parts do the jobs that cannot stop. A **central pattern generator** makes a rhythm with no rhythmic input; the **reticular formation** sets how awake an animal is.

## A lamprey's cord, cut from its brain



## What the hindbrain keeps running

- **breathing, heart rate,**  
blood pressure, swallowing
- **the reticular formation:**  
how awake, asleep or alert
- **one small cluster:**  
most of the brain's norepinephrine

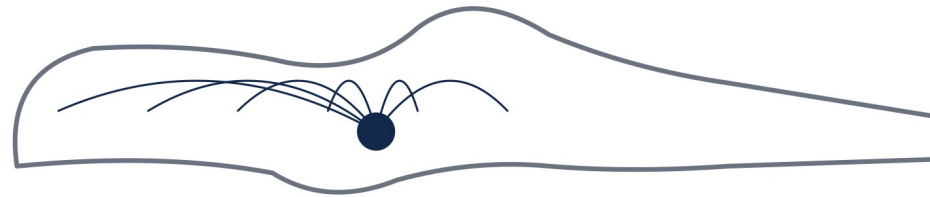
Down here, dedicated circuitry for a single behavior really does exist.

# The Midbrain: Orienting, and Where the Broadcasts Start

The tectum aims by finding a place on a map. And the broad projections of the last lecture start in a few small clusters.

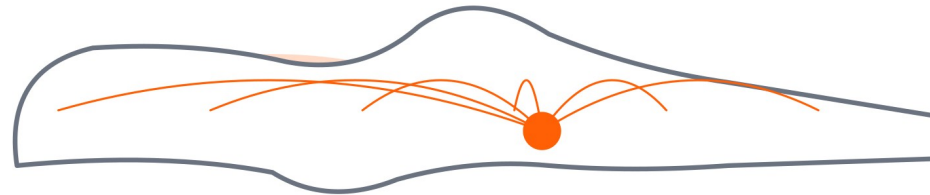
Three small places, and how far their fibers reach

**dopamine**  
a cluster in the midbrain



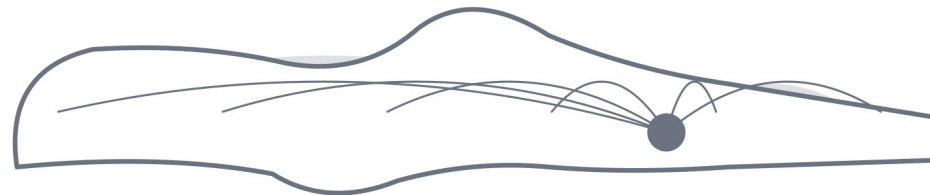
reaches  
the forebrain, mostly

**serotonin**  
cells along the midline



reaches  
most of the brain

**norepinephrine**  
one cluster, in the hindbrain



reaches  
most of the brain

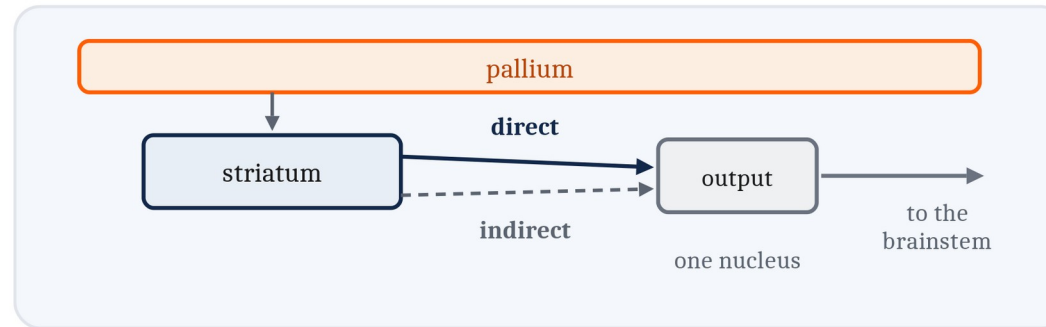
*the same brain three times; the shading is the territory each set of fibers reaches*

# The Forebrain's Old Core

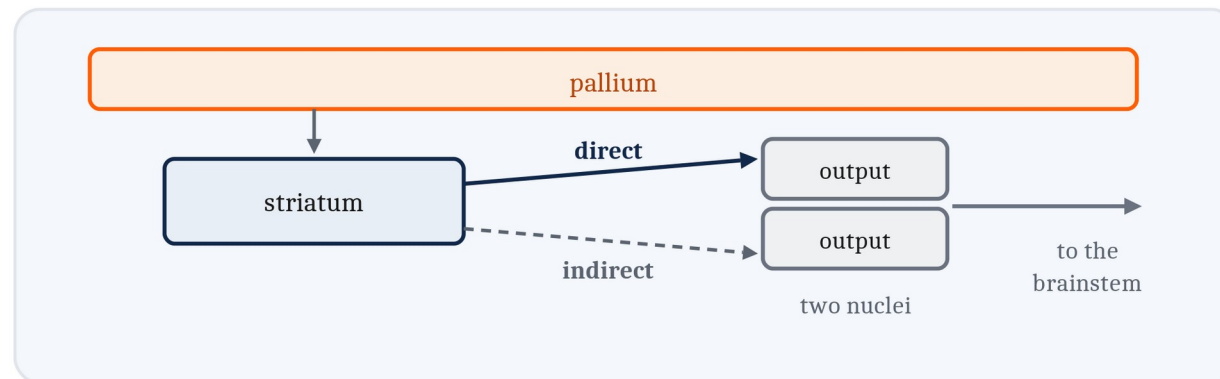
**Homeostasis** from the hypothalamus, selection from the basal ganglia, and valence from the **amygdala**, the last in jawed vertebrates.

The same selection circuit, twice

lamprey



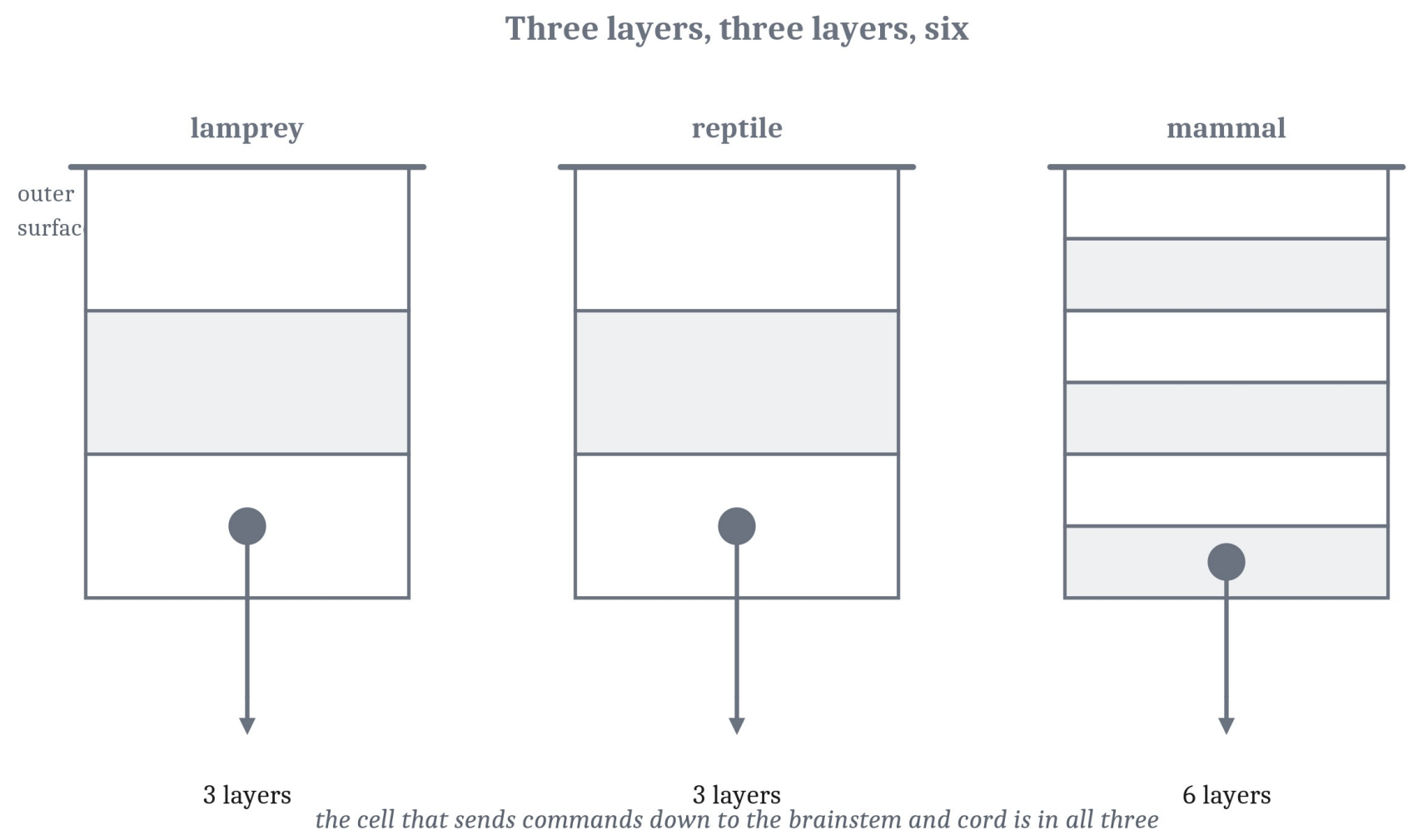
mouse



*same transmitters, same connections, same two routes — what differs is how many copies*

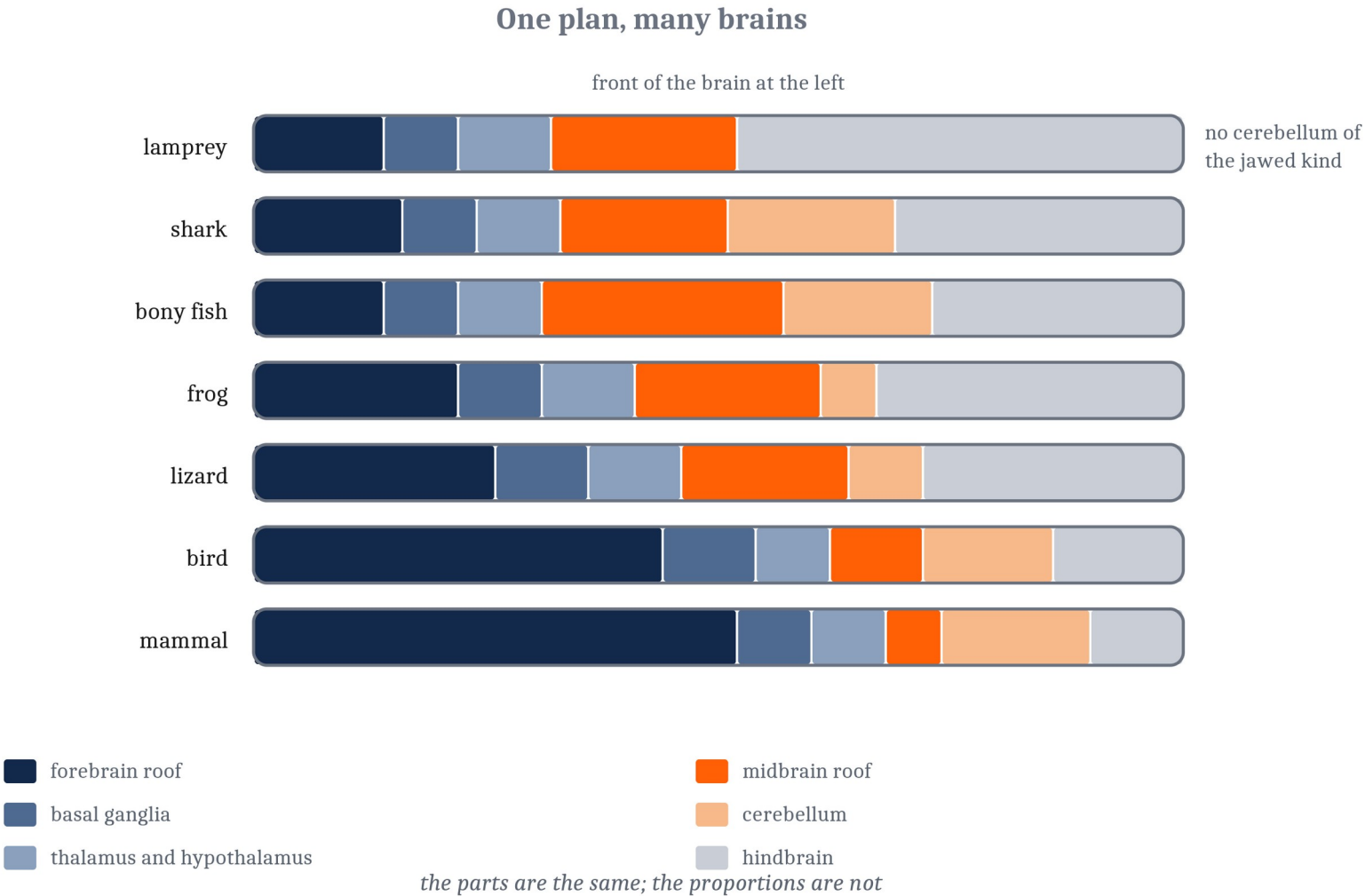
# The Pallium, and the Thalamus That Serves It

Every vertebrate has a pallium, built differently in each. Almost everything that reaches it passes through the thalamus.



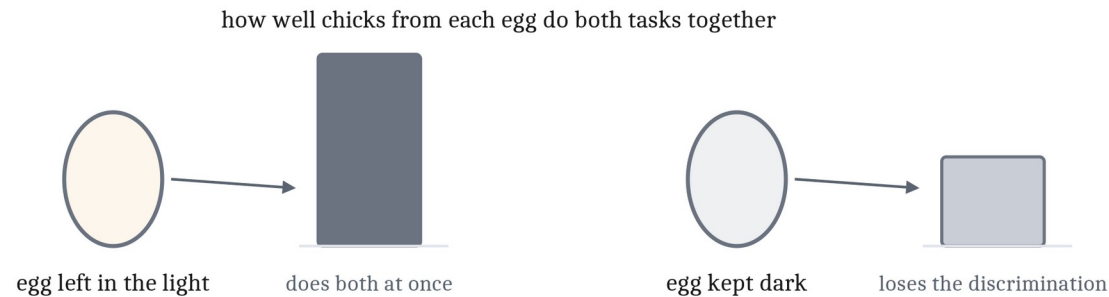
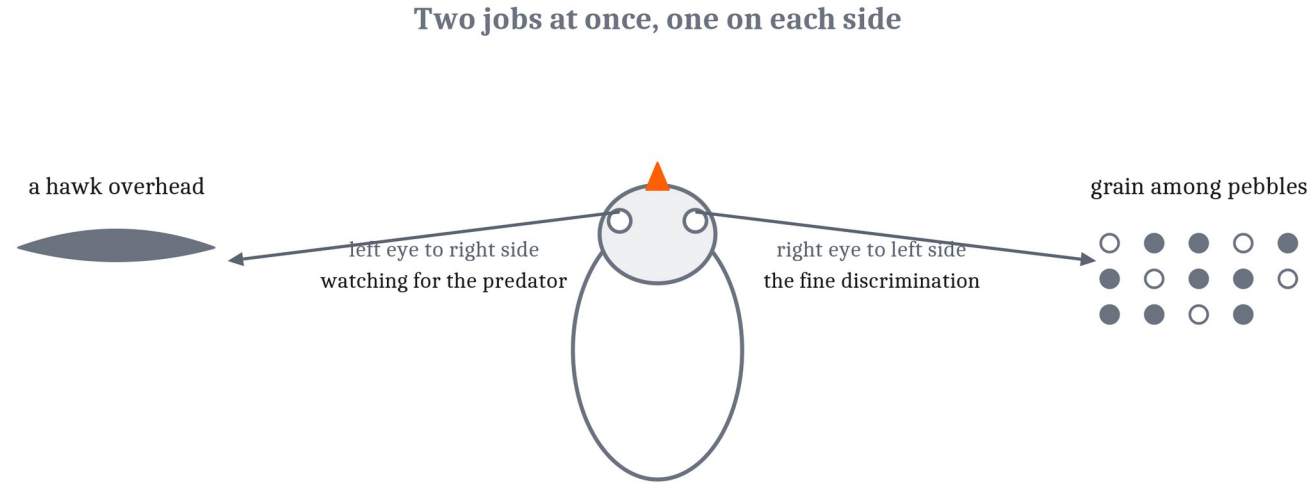
# One Plan, Elaborated Many Times Over

The same parts, in the same order, from lamprey to mammal. What changed is proportion and number, and it changed **in parallel, not in one climb**.



# Two Halves

**Lateralization:** the two sides of a brain doing different things. The chick shows why an individual benefits. Why a whole population leans the same way is a separate question.

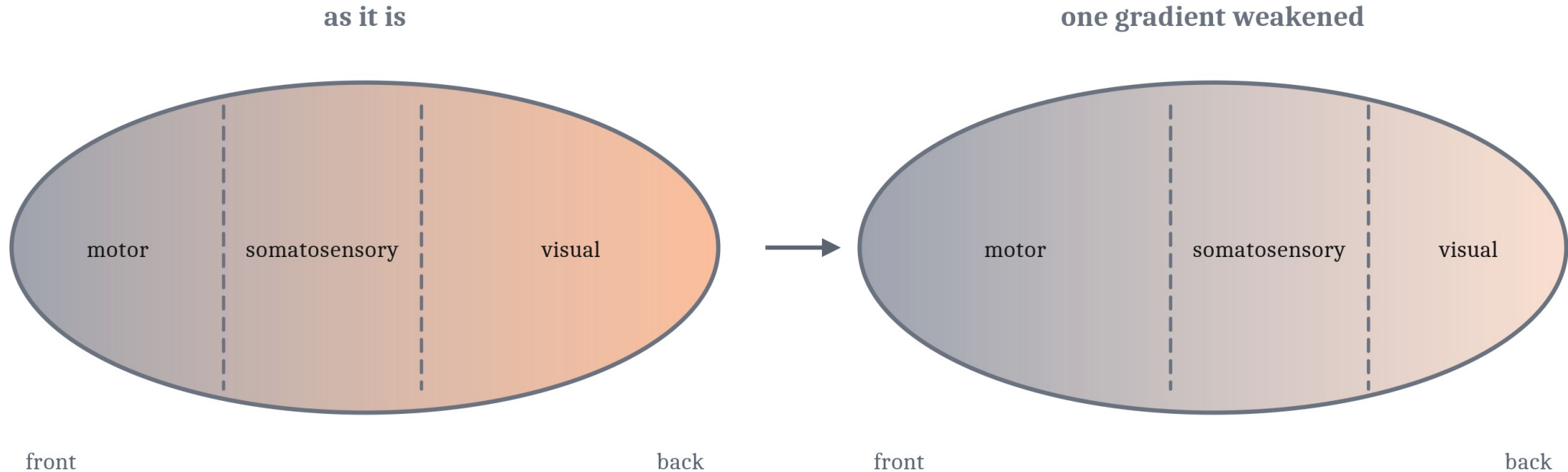


*light through the shell reaches the right eye only, and strengthens the asymmetry*

# Rules, Not a Blueprint

No genome lists which cell goes where. It holds a few rules run locally: read a gradient, listen to the neighbors, follow a cue, compete for a limited supply.

**Two gradients, and the areas that fall out of them**



where a cell sits in the pair of gradients sets which area it belongs to

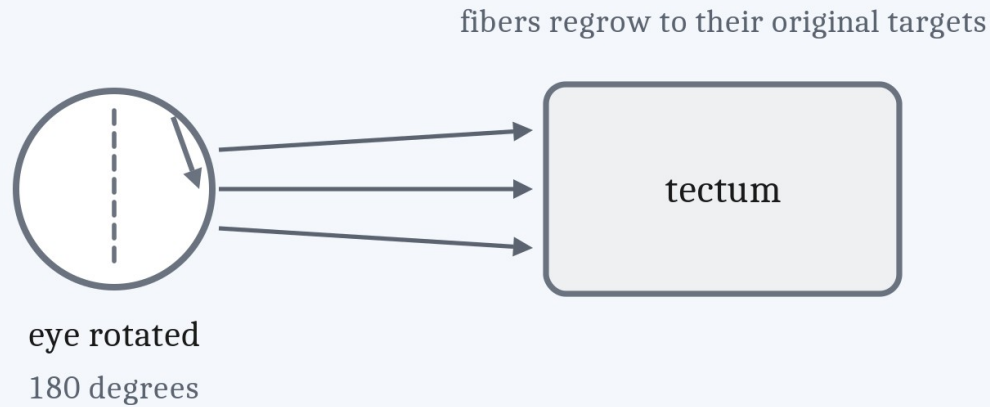
*shift one gradient and every boundary moves together*

# Made, Moved, Steered, and Culled

A neuron's birthday sets its layer. An axon finds its target by a position, not an address. Too many neurons are made, and the target settles which survive.

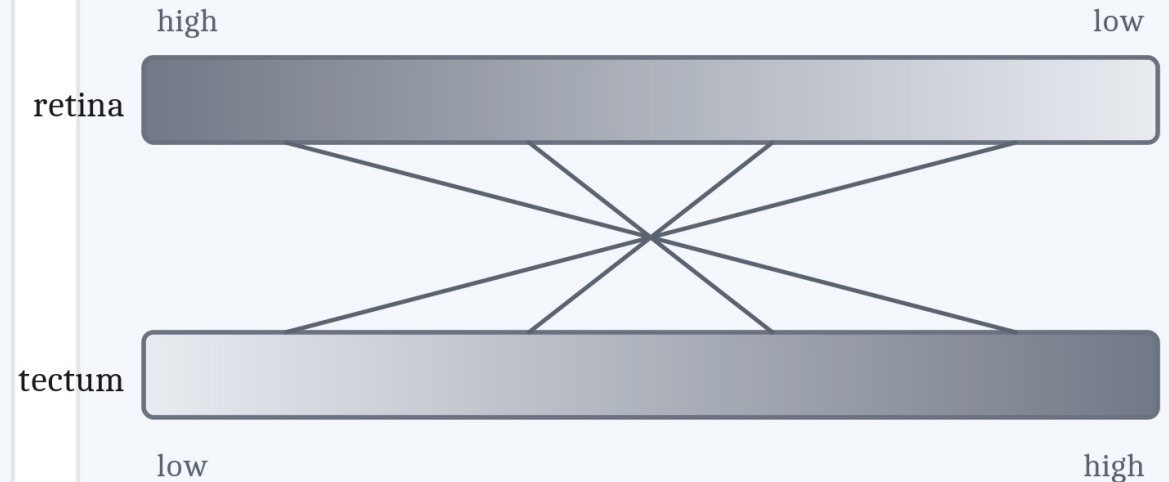
## Sperry's frog, and what was found instead of labels

### the experiment



so the frog strikes to the wrong side  
*a unique label per cell would explain that  
and so would something much cheaper*

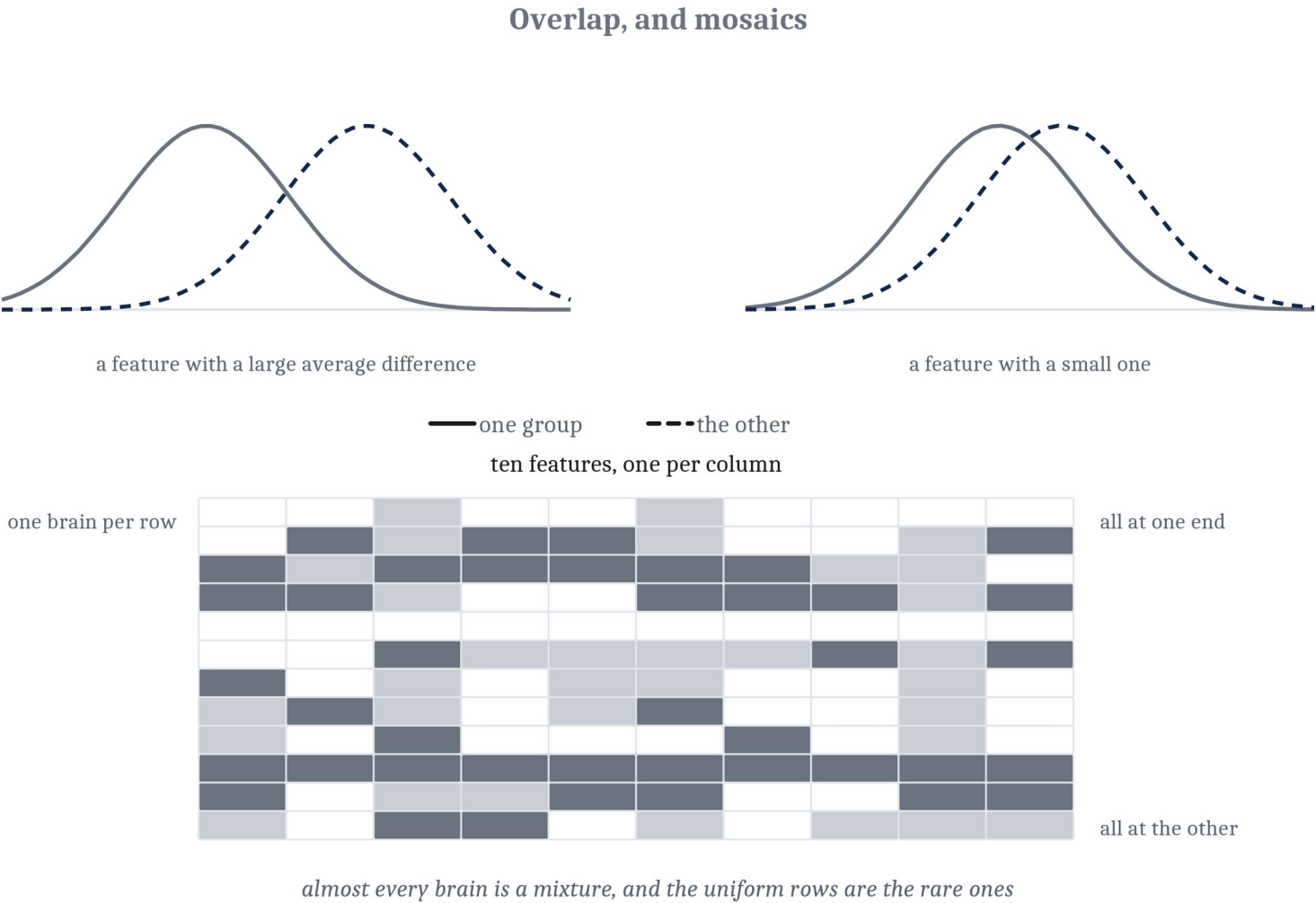
### what was found



each fiber ends where its level in one gradient  
matches the level in the other  
**a position, not an address**

# Sex Differences as Development

Each region responds to signals in its own way, so the mechanism predicts a **mosaic**, not two kinds of brain. Across more than fourteen hundred human brains, that is what was found.



# An Address Is Not an Account

Structures and functions map **many to many**. Neither list below is a list of separate functions done by separate parts.

## Structure

## Jobs it has been credited with

Amygdala

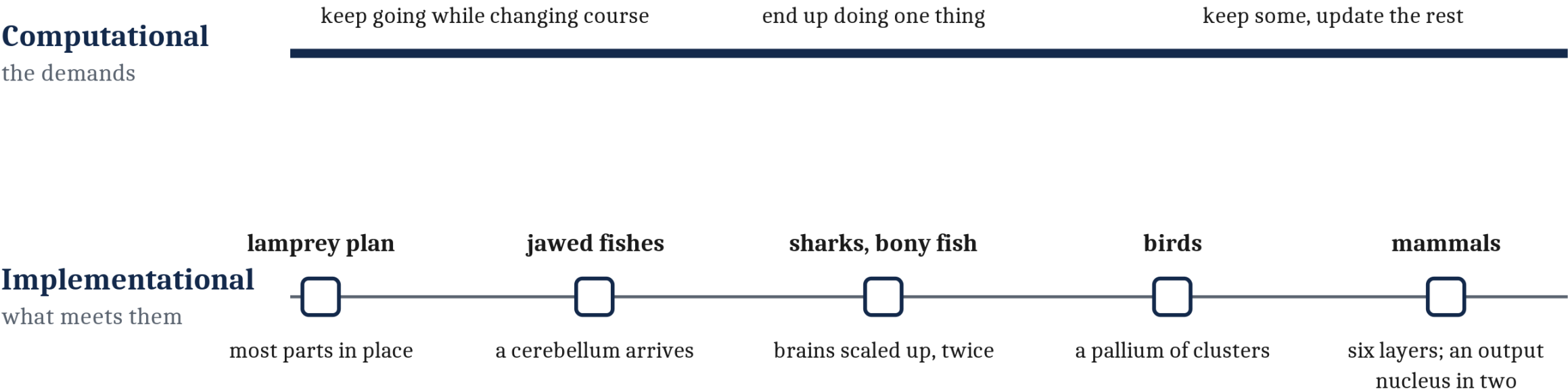
Learning that something predicts a threat; learning that something predicts a reward; responding to whatever is salient, good or bad; adjusting attention; influencing what is remembered

Striatum

Releasing one action and holding back others; learning which actions lead to reward; habits, as behavior becomes automatic; keeping track of how well things are going

# Old Problems, New Parts

The computational level barely moved. The implementational level was rebuilt again and again. So finding an arrangement does not tell us which demand it answers.



*elaborated many times over, in parallel, not climbed once toward mammals*

A problem can stand still while the machinery that answers it is remade several times over.

# Headlines About the Brain's "Centers"

**In your group, four minutes.**

Headlines regularly announce that a feeling, a trait or a habit has been found in one part of the brain.

Invent or recall one. What inference is it making?

What would a study have to show to earn the word *center*?

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